Rust - 0: Course intro

A language empowering everyone to build reliable and efficient software.

Who are we?

Henk Oordt

- Embedded software engineer
- 4 year Rust experience
- Work on IoT device using Rust
- Main contact



Who are we?

Folkert de Vries

- Embedded software engineer
- Write compiler in spare time with Rust
- 3 years of Rust experience



Who are we?

Ruben Nijveld

- Software engineer
- Rust for web and systems programming
- Introduced Rust at our company



What you'll learn

- Ability to write your own applications in Rust
  - Using popular crates (libraries)
- Ability to contribute to existing Rust projects
- The problems Rust aims to solve, and how
- Ability to judge whether Rust is for you

What you’ll learn

- Focus on practical, hands-on experience
- Focus on the 'why' of Rust and its features

Lecture format

- Interactivity in lectures and tutorials
- Activate prior knowledge
- Room for discussion
- Quiz questions

Course structure

- 9 modules (apart from this intro)
- 12 weeks

Course structure

Weeks 1 - 9

- 1 lecture
- 1 tutorial
- 1 set of exercises

Course structure

Weeks 10 - 12

- Lecture only
- Work on final project

Project intro

- A bigger project involving Rust
- Work in teams of 2
- Hand in proposal in **week 7 before lecture**
  - Any reparations due in **week 8 before lecture**
- Write small report (2-3 pages), to be handed in in **week 12 before lecture**
  - Introduction
  - Requirements
  - Design overview diagram
  - Design choices
  - Dependencies
  - Evaluation
- Present project in about 5-10 minutes in **week 12 during lecture/tutorial**

Tools you'll use

- Rustup
  - Installs `rustc` and `cargo`
- VS Code
  - Rust Analyzer plugin
  - Live share plugin
- Rust playground
- Rust 101 repo on GitHub

Today

Module A1

- Slightly longer lecture
- Start with exercises